

Maximizing Car Revenue For Drivers Through Payment Type



Agenda

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- Hypothesis Testing
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Problem Statement

In the rapidly evolving taxi booking industry, optimizing revenue is crucial for sustained success and driver satisfaction. Our objective is to leverage data-driven insights to enhance revenue streams for taxi drivers.

This research focuses on exploring the potential influence of payment methods on fare pricing, specifically examining the relationship between payment type and fare amount.

Research Question?

1. Is there a relationship between the total fare amount and payment type?
2. Can we encourage customers to choose payment methods that yield higher revenue for drivers without compromising the customer experience?

Data Overview

For this analysis, we employed the extensive dataset of NYC taxi trip records (2020-01). We conducted data cleaning and feature engineering to focus exclusively on the relevant columns necessary for our investigation.

Relevant Columns Used for This Research:

- passenger_count (1 to 5)
- payment_type (card or cash)
- fare_amount
- trip_distance (miles)
- trip_duration (minutes)

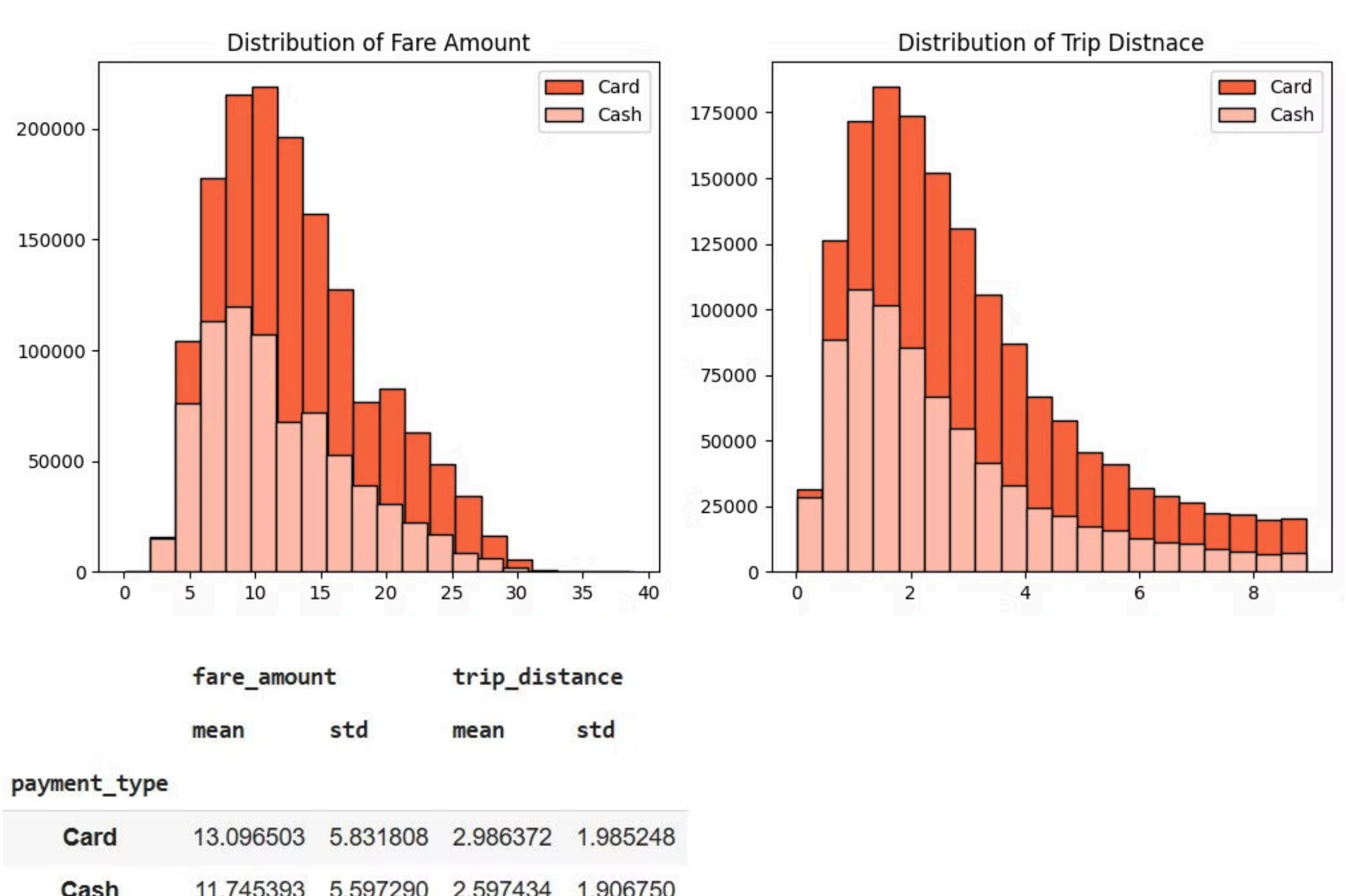
passenger_count	payment_type	fare_amount	trip_distance	trip_duration
5	Card	21.0	4.79	29.383333
2	Cash	5.0	0.81	4.066667
5	Cash	8.0	1.13	10.083333
2	Card	19.5	3.99	28.266667
2	Card	16.0	4.44	17.466667

Methodology

Steps	Description
Descriptive Analysis	Conducted statistical analysis to highlight key aspects of the data, with an emphasis on fare amounts and payment types.
Hypothesis Testing	Performed a T-test to assess the relationship between payment type and fare amount, testing the hypothesis that various payment methods affect fare amounts.
Regression Analysis	Applied linear regression to investigate the relationship between trip duration (calculated from pickup and drop-off times) and fare amount.

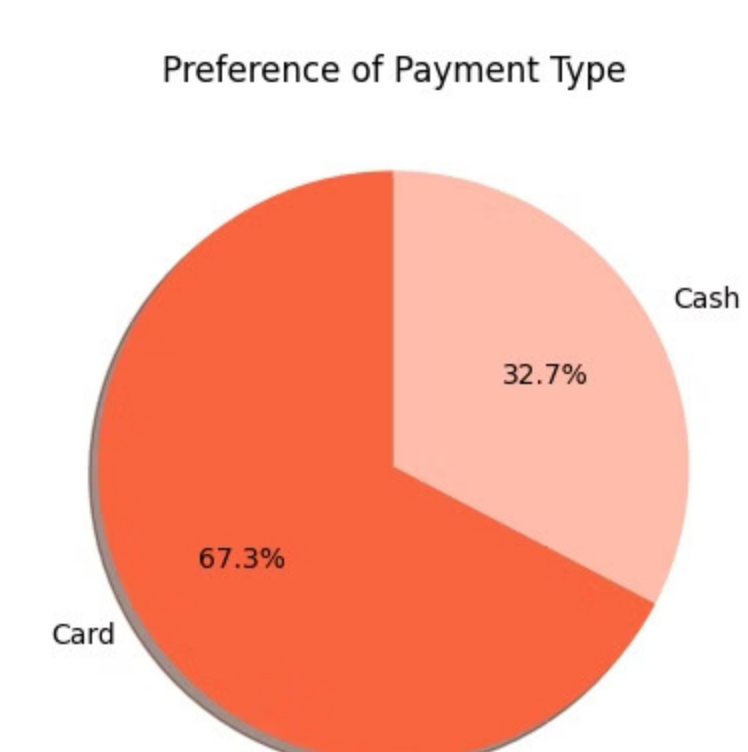
Journey Insights

- Customers using cards generally show a marginally higher average trip distance and fare amount than those paying with cash.
- This suggests that customers are more inclined to use cards for longer trips with higher fare amounts.



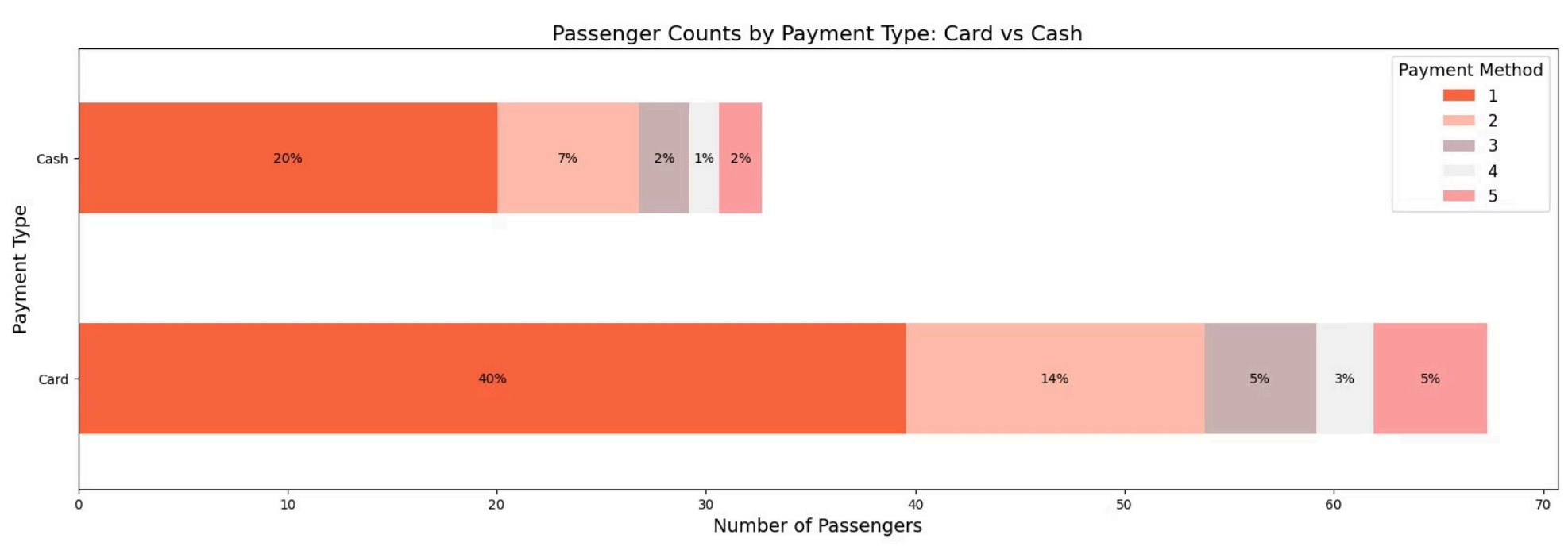
Payment Type Preferences

- The percentage of customers opting for card payments is notably higher than those choosing cash, with card transactions making up 67.5% of all payments, while cash payments account for 32.5%.
- This highlights a strong customer preference for card payments, likely driven by factors such as convenience, security, or incentives associated with card use.



Passenger Count Analysis

- Rides with a single passenger (passenger_count = 1) represent the largest share of card payments, accounting for 40% of all card transactions.
- Likewise, single-passenger rides dominate cash payments, comprising 20% of all cash transactions.
- There is a noticeable decline in transaction percentages as passenger counts increase, indicating that larger groups are less inclined to use taxis or may prefer alternative payment methods.
- These insights underscore the importance of examining both payment method and passenger count when analyzing transaction data, as they reveal valuable information about customer behavior and preferences.



Hypothesis Testing

- Null Hypothesis: There is no difference in the average fare between customers using credit cards and those using cash.
- Alternative Hypothesis: There is a difference in the average fare between customers using credit cards and those using cash.

With a T-statistic of 169.19 and a P-value of less than 0.05, we reject the null hypothesis. This indicates a significant difference in average fare between the two payment methods.

Recommendations

- Encourage customers to use credit cards to maximize revenue potential for taxi drivers.
- Implement strategies like offering incentives or discounts for credit card transactions to motivate customers to select this payment method.
- Additionally, provide seamless and secure credit card payment options to enhance customer convenience and promote the adoption of this preferred payment method.

Thank You